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Abstract

Many properties used to study and classify astronomical objects depend upon the opacities of their constituent plasmas. Spectral line shapes are essential for determining the temperature and density of observed stellar and compact objects and understanding and accurately modeling plasma opacities is crucial for computational stellar astrophysics. However, there are discrepancies between observed and calculated line shapes in the high energy density (HED) regime. We explore an application of molecular dynamics to line shape models to ascertain potential sources of these discrepancies.

Plasma

A plasma is a partially- or fully-ionized gas found in a variety of places.

High energy density (HED) plasmas are found in stars and compact objects, such as white dwarfs (WDs).



Stellar Spectra

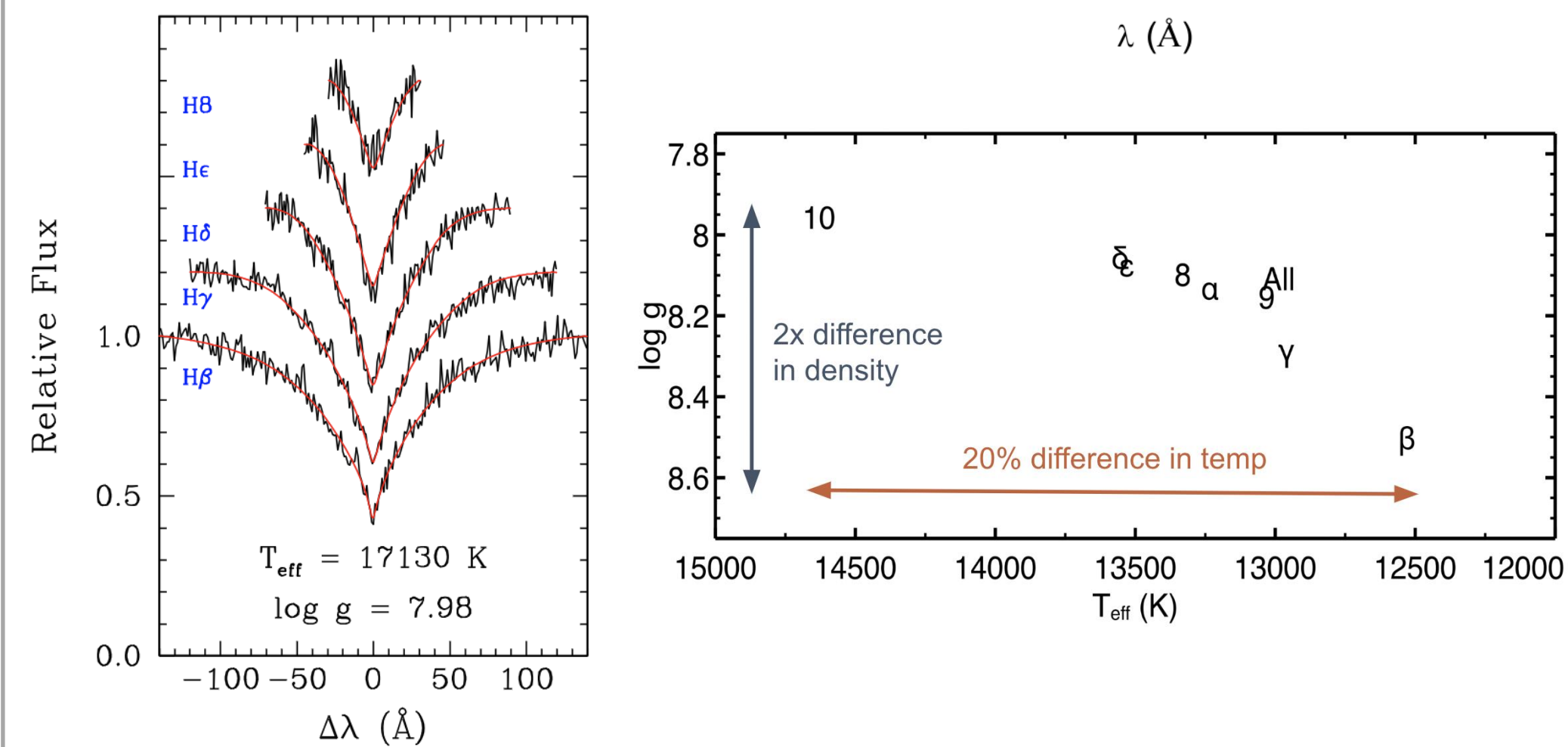
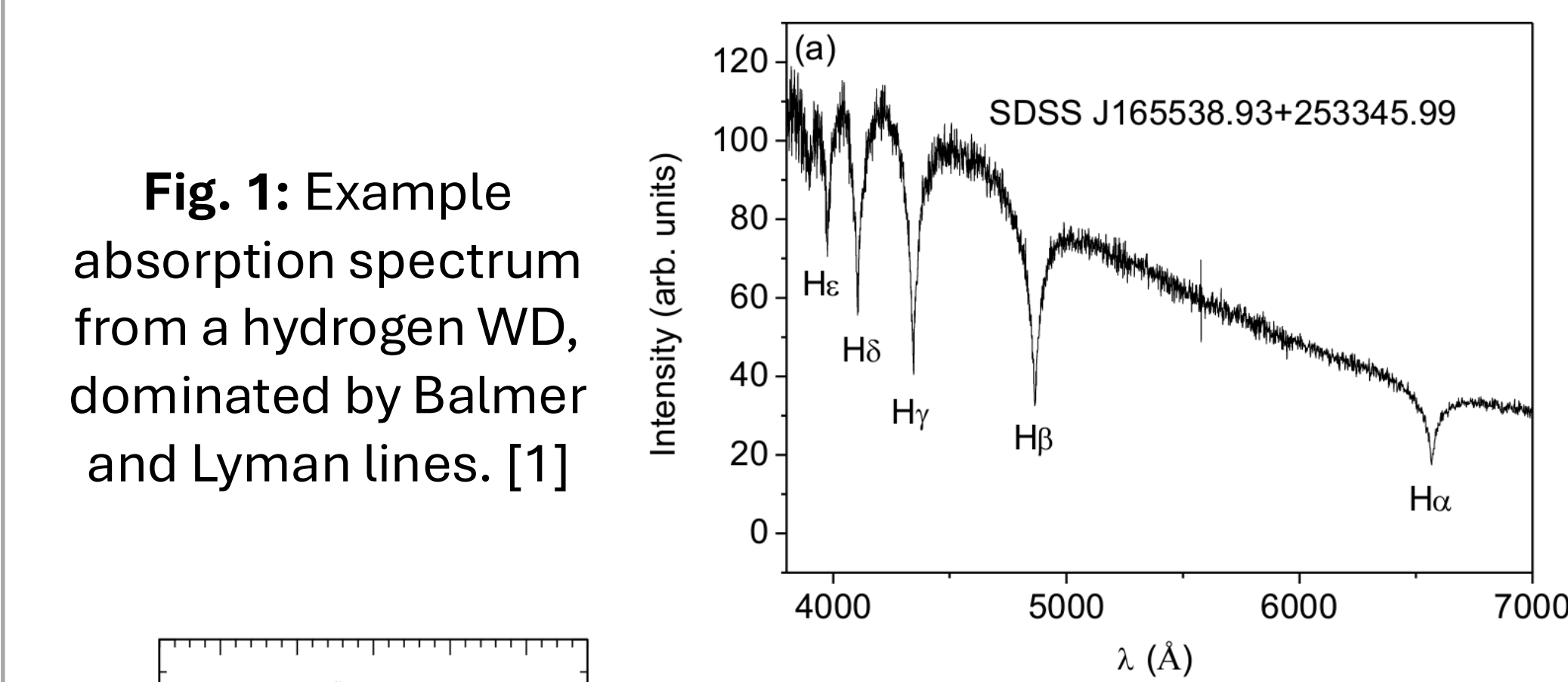


Fig. 2: Full radiation transport fits to each line of Fig. 1 for a single star [2, left] and their corresponding temperature and density measurements [3, right].

Opacity Line Shapes

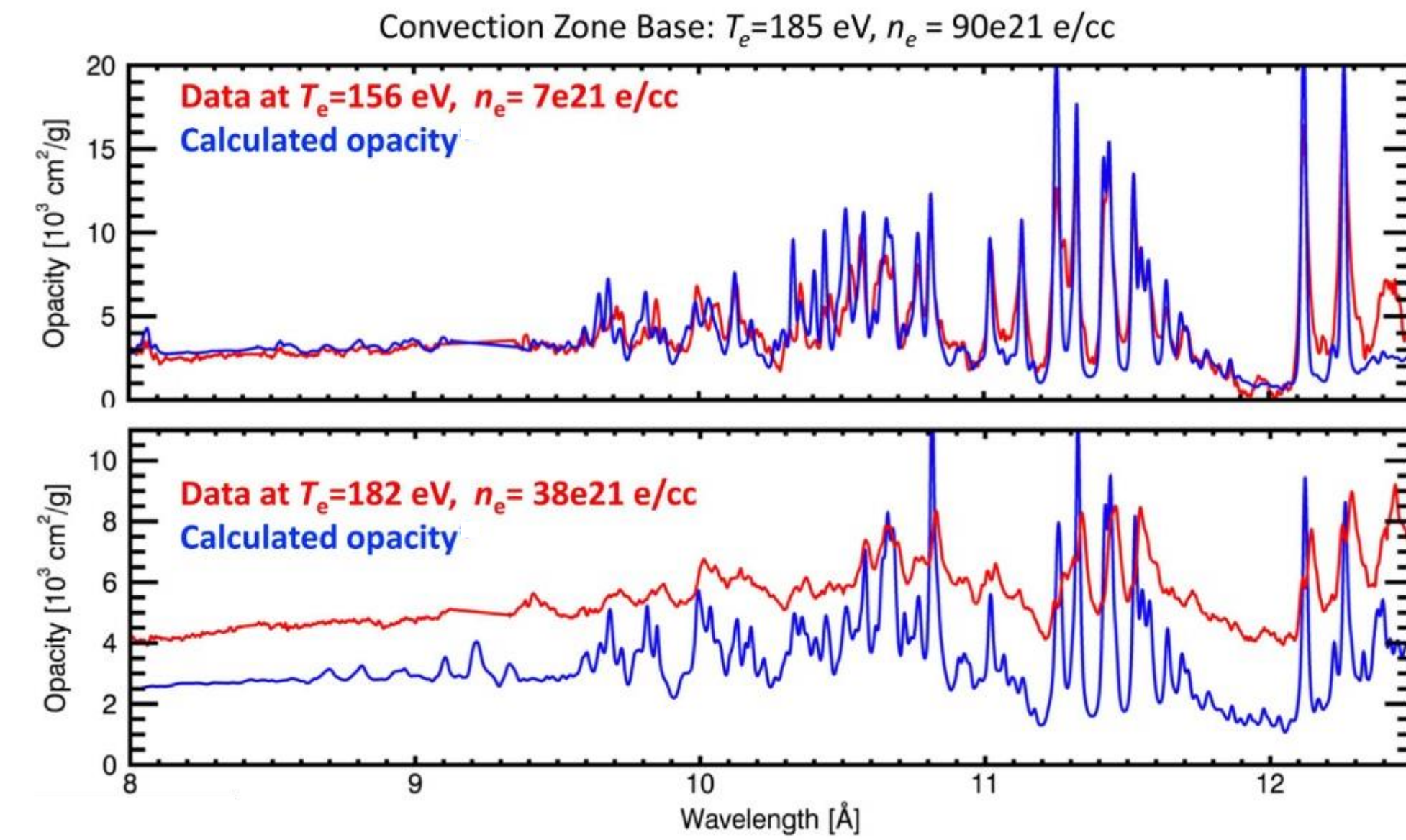


Fig. 3: Iron plasma opacity data (red) and calculations (blue) at two temperatures and densities, showing a discrepancy in both the bound-bound (horizontal lines) and bound free (spectral lines) transitions. [4]

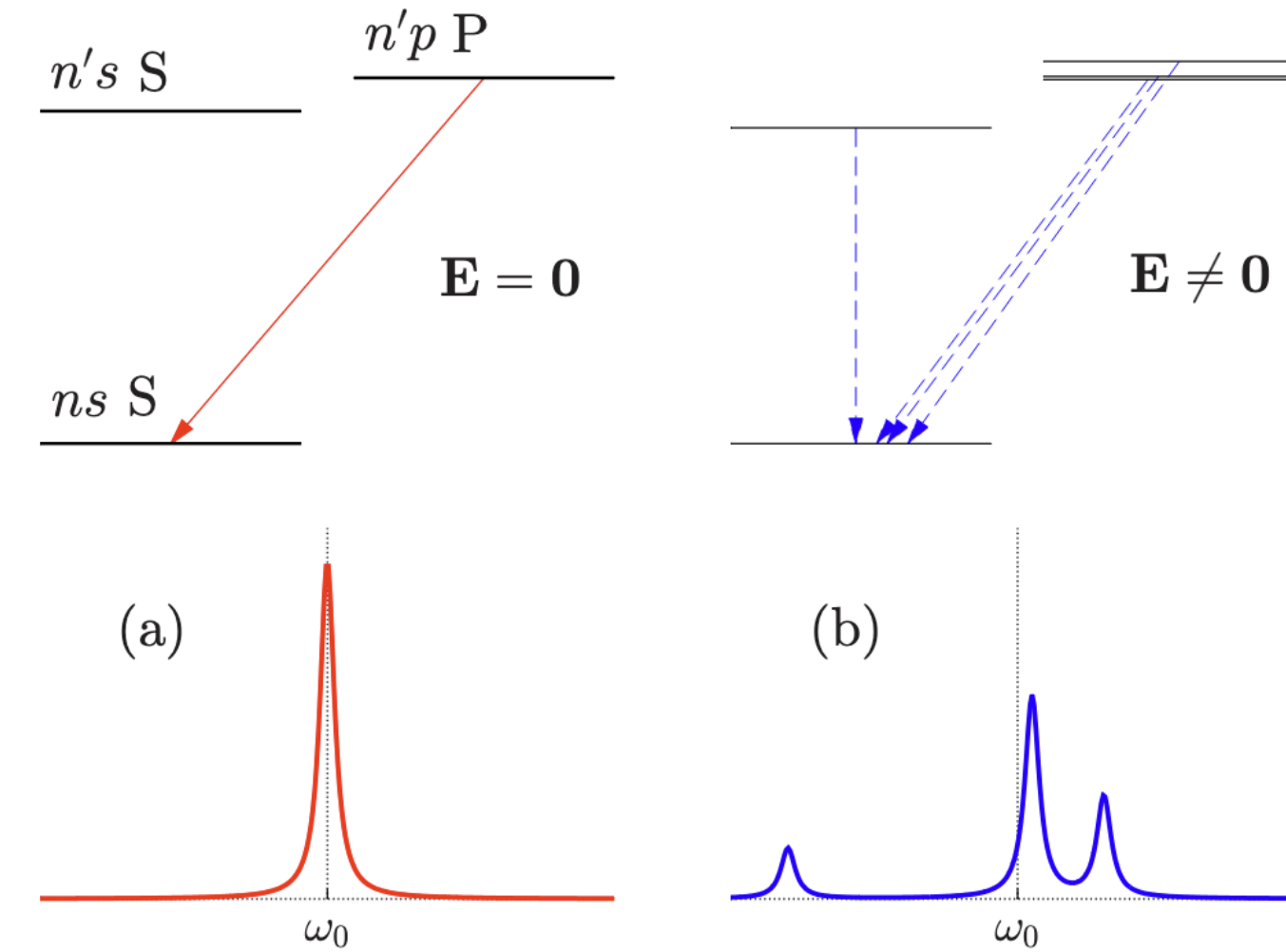


Fig. 4: Stark broadening: an example of how electric fields influence line shapes. [5]

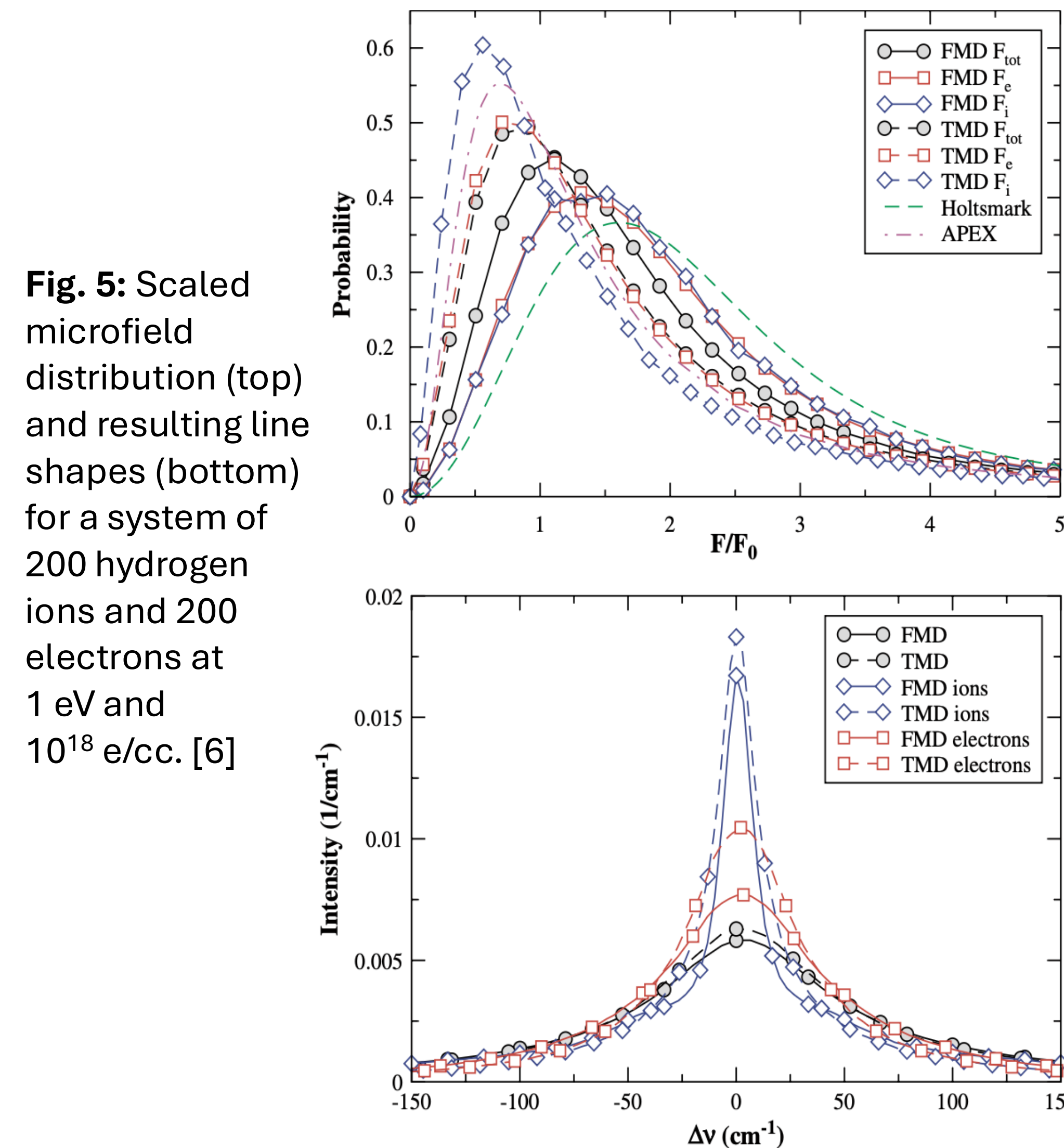


Fig. 5: Scaled microfield distribution (top) and resulting line shapes (bottom) for a system of 200 hydrogen ions and 200 electrons at 1 eV and 10^{18} e/cc. [6]

Molecular Dynamics (MD) for Microfields

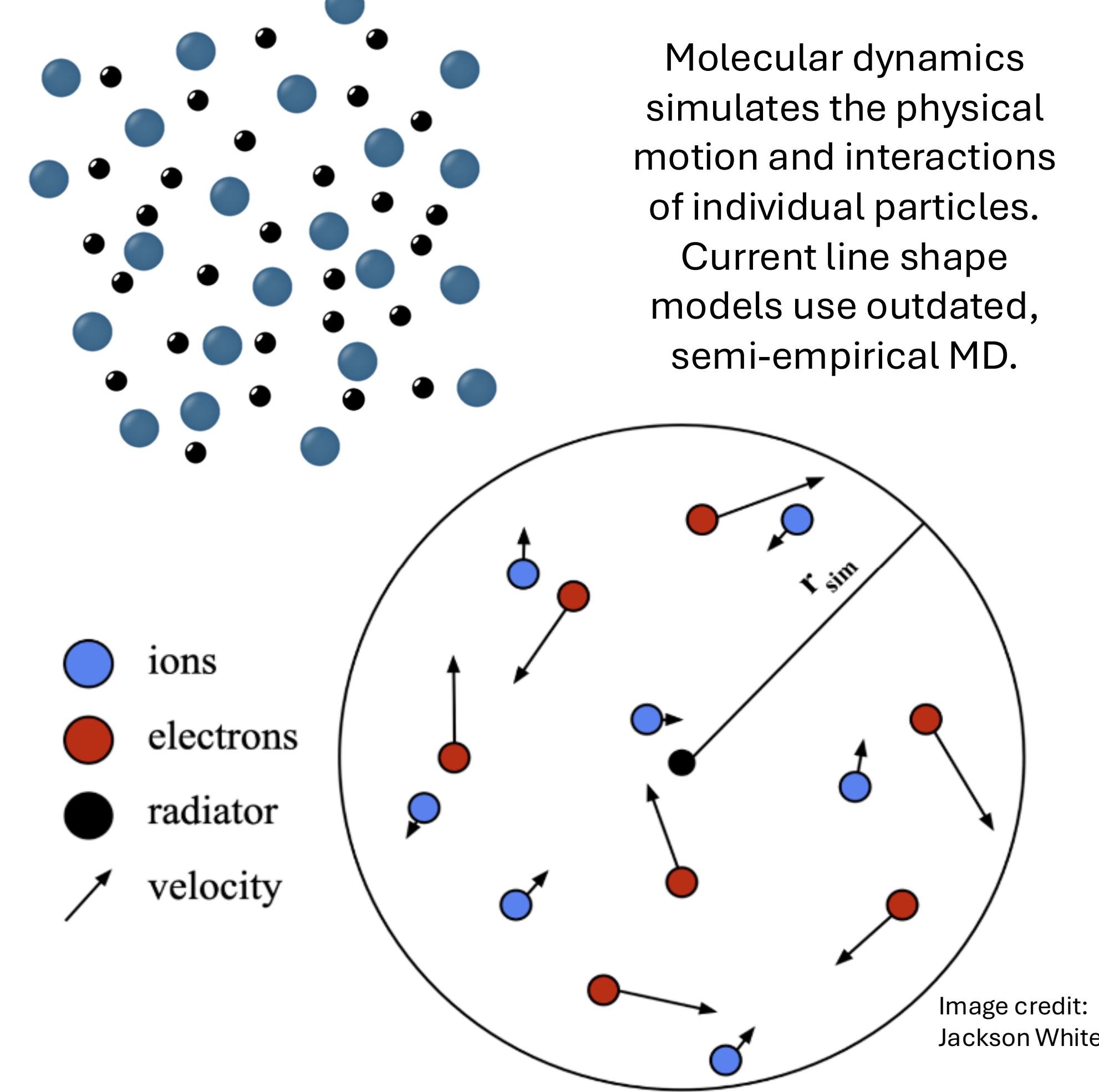


Fig. 6: An example of the current dynamical model used in the line broadening program Xenomorph [7] (Unrelated to Fig. 3).

Fig. 7: LAMMPS [8] is the molecular dynamics code we have chosen to produce microfield distributions for Xenomorph.

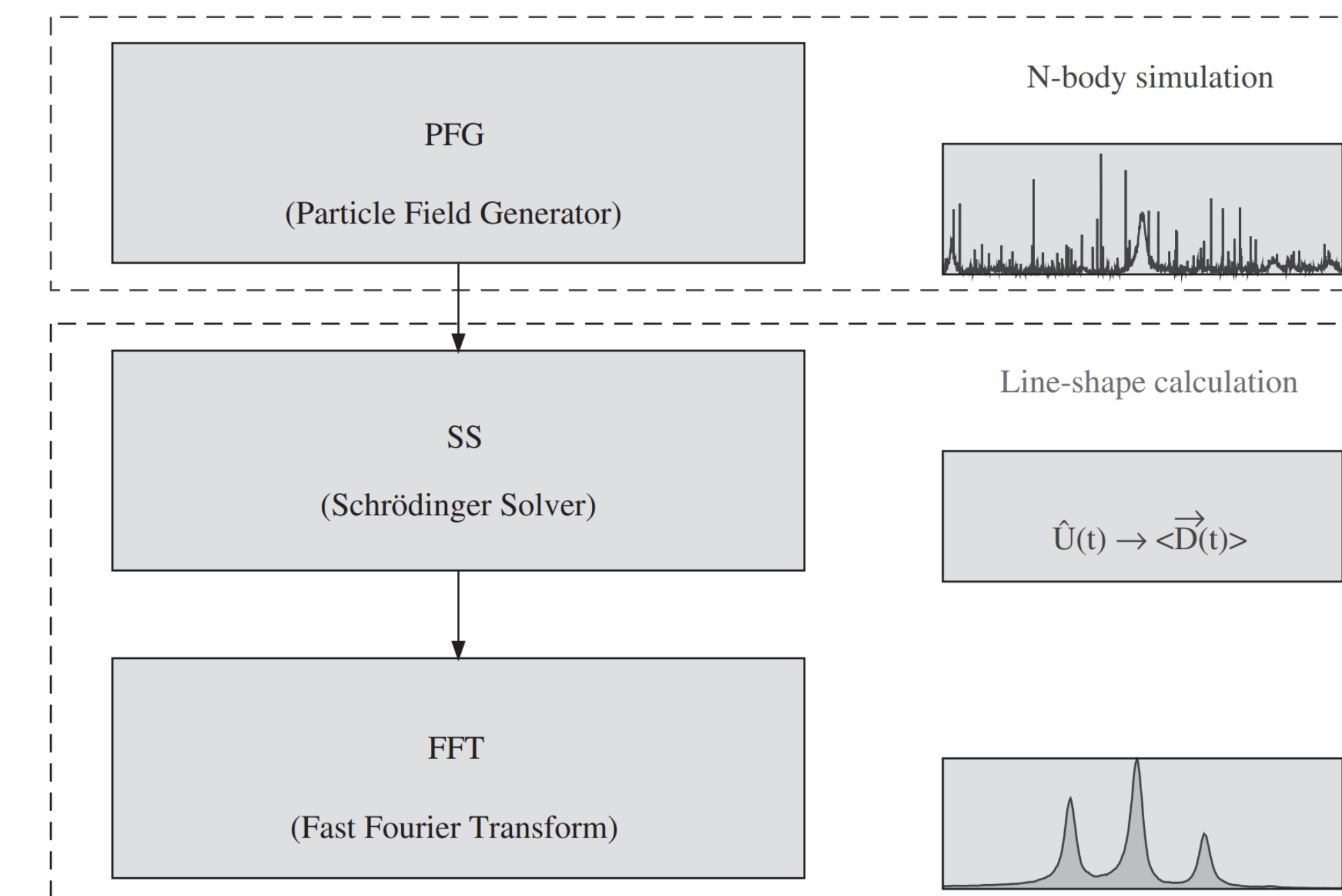
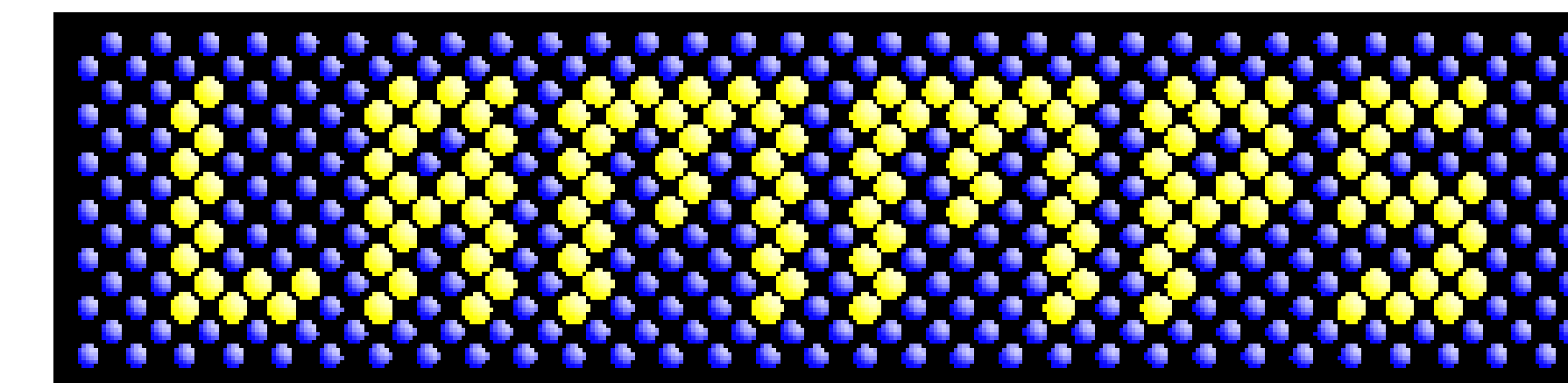


Fig. 8: A diagram showing how microfield distributions are converted into line shapes, including the following equation. [9]

$$I(\omega) = \text{Re} \left[\frac{1}{\pi} \int_0^\infty e^{i\omega t} \text{tr} \left(\vec{D} \cdot \vec{D}(t) \rho \right) dt \right]$$

Preliminary Results

Fig. 7: A comparison between the attractive quantum statistical potential [10] we use to correct LAMMPS at close-approach, the Kelbg equation that it corrects, and a pure-Coulomb potential at 1 eV and 10^{18} e/cc.

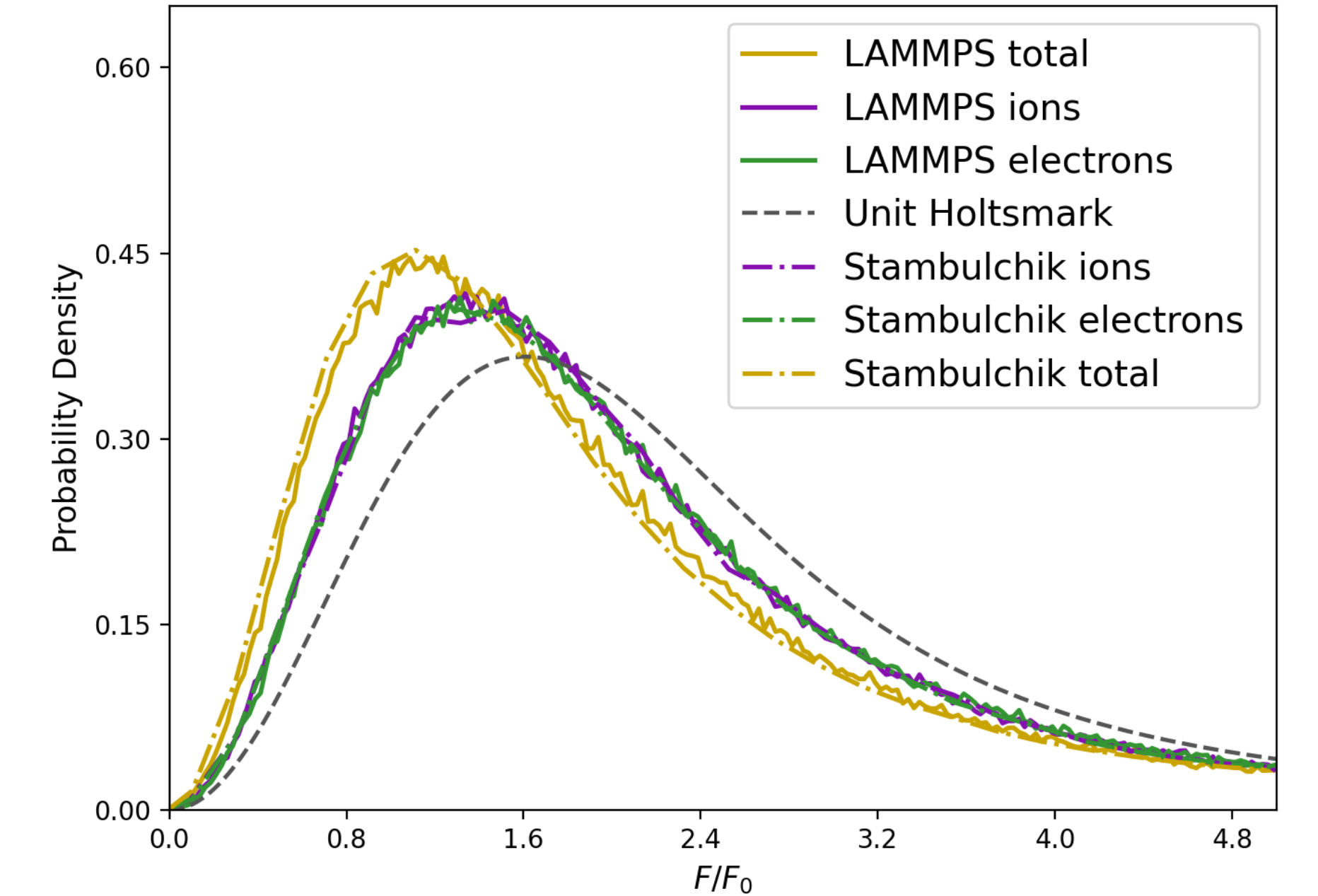
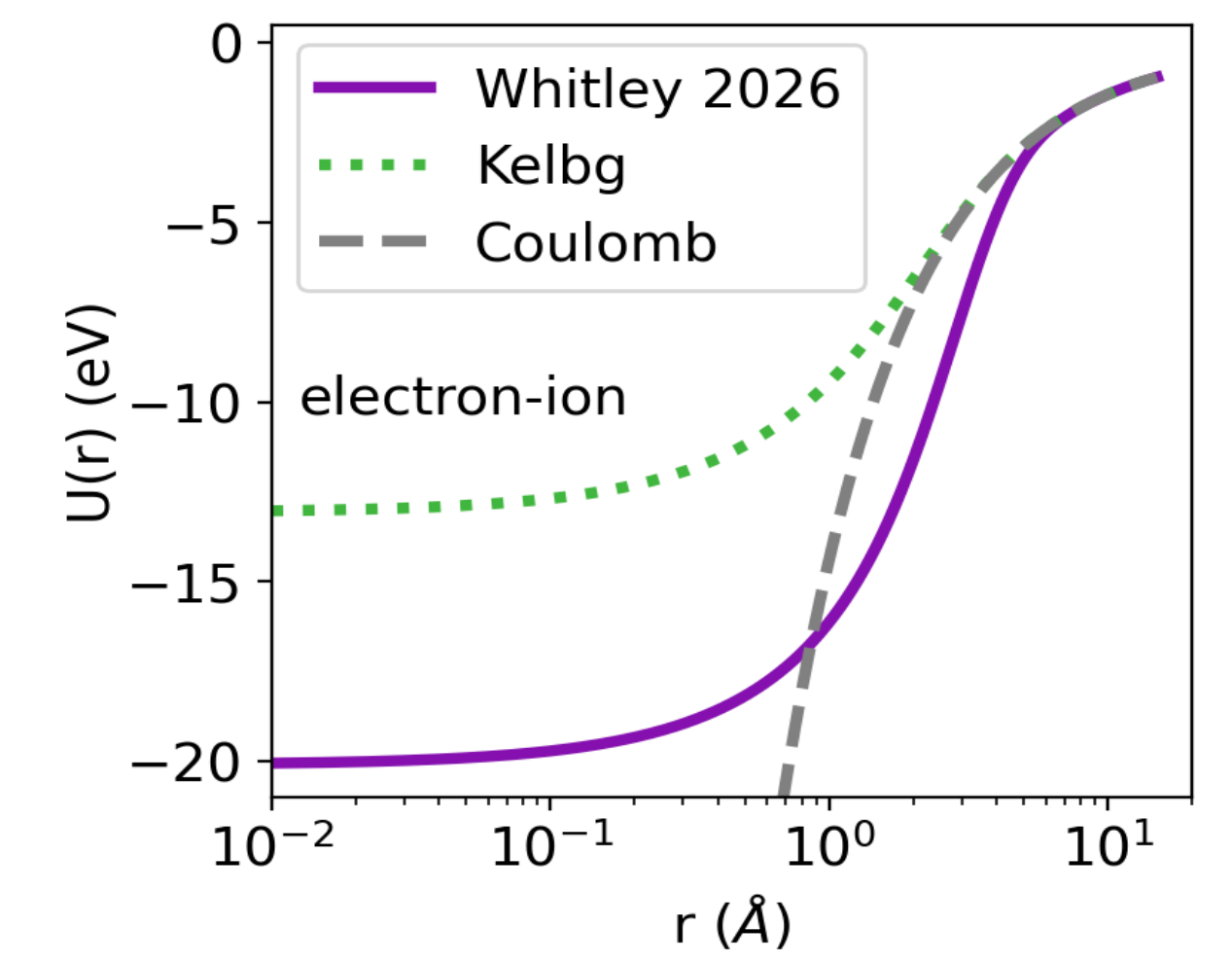


Fig. 8: Preliminary microfield distributions at the target conditions. The electron and ion microfields match those of Fig. 5 very well, but our total microfields are slightly skewed.

Future Work

- Improve model fit and continue debugging LAMMPS
- Simulate additional conditions
 - Higher temperatures
 - Higher densities
 - Larger Z-values
- Apply to Xenomorph to determine if improved molecular dynamics mitigates the iron opacity discrepancy in Fig. 3

References

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